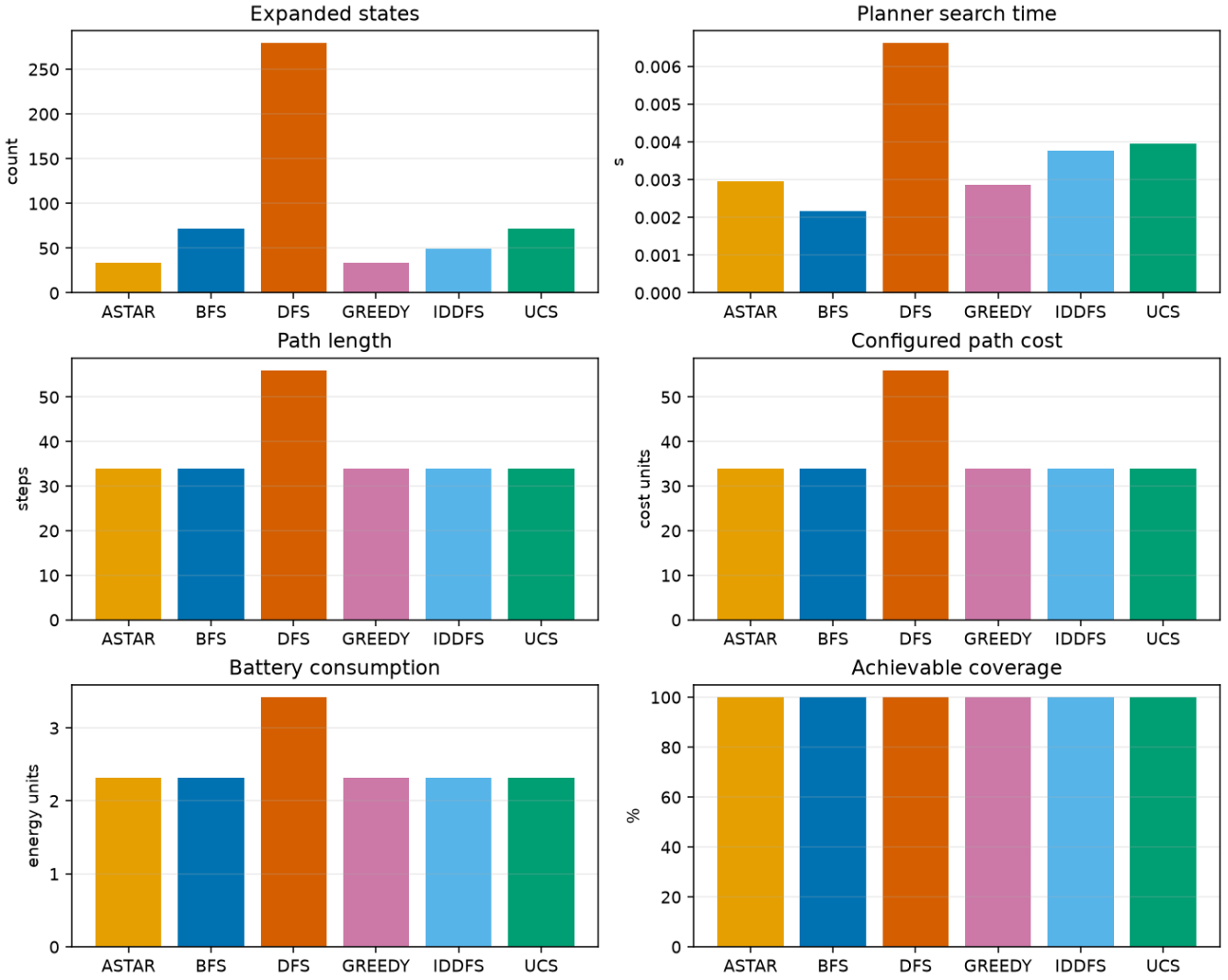
Coverage comparison

Runs share map, robot, policy, neighbor order, cost model, heuristic, and mission mode.

Algorithm	n	Expanded	Search median (s)	IQR (s)	Steps	Cost	Energy	Coverage %	Stop
ASTAR	5	34.0	0.002966900006867945	8140034567421347	34.0	34.0	2.3199999999999978	100.0	mission_complete
BFS	5	72.0	0.002178000402638265	3620037068470342	34.0	34.0	2.3199999999999978	100.0	mission_complete
DFS	5	279.0	0.006614200072360196	8230052370074591	59.0	59.0	3.4199999999997175	100.0	mission_complete
GREEDY	5	34.0	0.002871299511750221	1099341213763743	34.0	34.0	2.3199999999999978	100.0	mission_complete
IDDFS	5	49.0	0.003769000410102308	6460034787651458	34.0	34.0	2.3199999999999978	100.0	mission_complete
UCS	5	72.0	0.003962400369340614	2700034013793407	34.0	34.0	2.3199999999999978	100.0	mission_complete



Assumptions and limitations

Map grid: 6 × 6; resolution 1.0 m/cell. Movement: four-neighbor-cardinal; neighbor order: ['up', 'right', 'down', 'left']; heuristic: manhattan; cost model: {'transition': 'unit'}; tie-breaking: recorded algorithm default; coverage policy: {'id': 'nearest_uncleaned_manhattan_v1', 'configuration': {}}.

A single scenario does not support a generalized superiority claim. DFS depends on neighbor order. Greedy search is not guaranteed to return an optimal path. IDDFS may repeat expansions across depth limits. Simulated battery values are abstract model outputs, not real appliance measurements.